Shafkat Haque

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Objective

Dedicated Electrical Engineering MEng student specializing in Energy and Environment, with hands-on experience in power systems, renewable energy integration, and electrical system analysis. Proficient in ETAP simulation, OpenSolar PV design, load flow analysis, and technical documentation. Seeking an opportunity to contribute analytical and engineering expertise toward sustainable, high-impact energy solutions.

Education

Schulich School of Engineering, University of Calgary Calgary, AB

Master of Engineering, Electrical and Computer Engineering; Specialization: Energy and Environment *Expected Dec. 2026*

BRAC University Bangladesh

Bachelor of Science, Electrical and Electronics Engineering *Aug. 2022*

Core Skills

Electrical Engineering Software: ETAP, OpenSolar, AutoCAD, Revit, SolidWorks, Cadence, PSpice, Arduino IDE, MATLAB
Data Analysis & PCB Design: Power BI, Advanced Excel, PCB and Schematic Design
Programming: Python (NumPy, Pandas, Scikit-learn)
Engineering Expertise: Power System Analysis, Load Flow Analysis, Grid Integration Studies, Technical Documentation, Electrical System Modeling, Renewable Energy Systems, HSE Compliance, Electrical Safety Standards
Office & Project Tools: Microsoft Office Suite (PowerPoint, Word, Excel, Access), MS Project

Experience

Digital Marketing Executive Oct. 2022 – Mar. 2024

DigiFea *Dhaka, BD*

- Planned and executed search engine optimization (SEO) strategies, including keyword research, on-page optimization, technical audits, and link-building, to improve organic rankings and website traffic for client businesses.
- Created, launched, and optimized Google Ads campaigns (Search, Display, and Remarketing), managing budgets, bid strategies, ad copy, and A/B testing to maximize ROI and reduce cost-per-acquisition.
- Communicated with international clients via email, Zoom, Skype, and project management tools to gather requirements, provide updates, and align campaigns with business objectives.

Projects

Advanced Grid-Integrated Solar PV System Design and Impact Analysis Apr. 2026

- Developed and simulated a grid-integrated solar PV system using ETAP, performing load flow analysis, time-series simulation, reverse power flow studies, voltage regulation assessment, and N-1 contingency analysis for commercial electrical infrastructure.
- Designed and optimized a 363.55 kW rooftop photovoltaic system in OpenSolar, including PV layout development, string sizing, MPPT configuration, inverter selection, shading analysis, and annual energy yield estimation using Calgary-specific irradiance data.
- Produced detailed AutoCAD electrical single-line diagrams (SLD) for PV interconnection, inverter feeder circuits, protection devices, and utility integration in compliance with Canadian Electrical Code (CEC) and industry engineering standards.
- Integrated engineering analysis, technical validation, and performance modeling using ETAP, OpenSolar, AutoCAD, Arduino IDE, and Microsoft Office Suite to evaluate system reliability, energy efficiency, smart inverter control performance, and economic feasibility for large-scale renewable energy deployment.

Frequency Stability with High Renewable Penetration Dec. 2025

- Performed frequency-stability modeling by calculating inertia constants, RoCoF, and stability indices to assess renewable-rich grid performance.
- Evaluated a 45 MW disturbance to determine UFLS thresholds and system recovery characteristics.

- Compared NREL, ERCOT, and IEEE methodologies to synthesize key operational and modeling insights for low-inertia systems.
- Analyzed applicability of grid-forming inverters, synthetic inertia, and BESS for improving dynamic frequency support.

Optimizing Energy Consumption in Smart Buildings – A Machine Learning Approach Apr. 2025

- Devised and deployed machine learning models (Linear Regression, KNN, Random Forest) to predict real-time energy demand in commercial buildings.
- Engineered and merged multi-source datasets (building metadata & weather data) to generate feature-rich data pipelines and derived variables such as HVAC, lighting, and occupancy proxies for enhanced model accuracy.
- Implemented data preprocessing pipelines using Scikit-learn's ColumnTransformer, integrating StandardScaler and OneHotEncoder for automated numeric and categorical transformations.
- Formulated learning and validation curves to analyze model complexity and prevent overfitting, ensuring reliability across unseen building data.

Smart Water Quality Monitoring and Purification System Development Dec. 2021

- Created an efficient water purification system using UV lights.
- Monitored water quality using PH sensor, Turbidity sensor, TDS meter, and Ultrasonic sensor.
- Programmed Arduino microcontroller with Python to automate water quality monitoring.
- Purified water using Sediment (PP), Gas Filter (GAC), Black Carbon (CTO), Membrane, and Post Carbon Filter.

Voice-Controlled Robot Jun. 2020

- Developed a voice-controlled robot using Arduino Uno and embedded C, enabling speech-based motion control.
- Designed, soldered, and integrated microphone, motor driver, and sensor modules to build a stable hardware control system.
- Improved response accuracy by optimizing command-recognition logic and motor-control routines.

Volunteer Experience

Volunteer of Awards Committee & Health and Dental Service Standard Committee May 2025
Graduate Students' Association (GSA), University of Calgary *Calgary, AB*

- Optimized plans for student welfare with committee members.
- Contributed to the panel by fulfilling assigned works and tasks.

Assistant Director in Electrical and Electronics Club Jan. 2021
EEC, Brac University *Dhaka,BD*

- Assisted in planning and executing technical workshops, seminars, and industry talks.
- Coordinated circuit design contests, robotics competitions, and project showcases.

Certifications and License

Valid Class 5 Alberta's License
WHMIS 2015 and First Aid – Scheduled for completion in December 2026